Modelling with Functions

Exam code: 9MA0

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Updated on 19 June 2025

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Modelling with functions

What is a mathematical model?

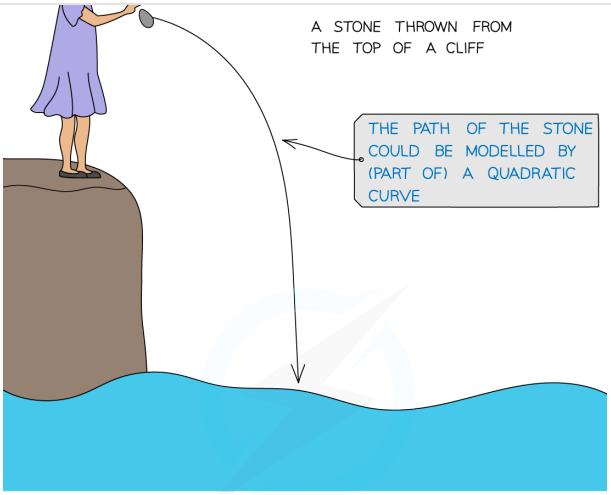
- A **mathematical model** simplifies a real-world situation so it can be described using mathematics which can then be used to make predictions
  - The path a stone will take if thrown from the top of a cliff
  - The number of toys a factory can produce in a day
- **Assumptions** about the situation are made in order to simplify the mathematics
  - Air resistance on the stone can be ignored
  - The machines/people at the factory produce toys at a constant rate
- Models can be **refined** (improved) if further information is available or the model is compared to real-world data
  - The mass of the stone needs to be considered
  - 30-minutes downtime per day is allowed for machine repairs/maintenance

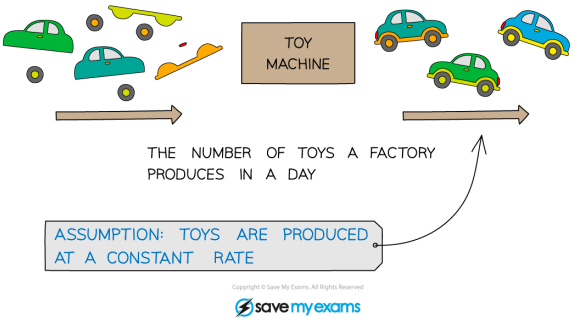
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it





How do I use mathematical models?

- There will be no one-size-fits-all step-by-step guide to solving modelling questions
- A combination of skills and **problem-solving** skills will be needed

FOLLOWING A STEADY RISE IN SALES OF A PARTICULAR PRODUCT OVER THE PREVIOUS YEAR A SHOP MANAGER SUGGESTS A LINEAR MODEL OF THE FORM $N=64+4m$ SHOULD BE USED TO PREDICT FUTURE SALES OF THE PRODUCT WHERE N IS THE NUMBER OF PRODUCTS SOLD AND t IS THE NUMBER OF MONTHS AFTER 1 JANUARY 2020.

a) HOW MANY PRODUCTS ARE PREDICTED TO BE SOLD **DURING** JANUARY?

b) WHAT IS THE MEANING OF THE VALUE 4 IN THE MODEL?

c) HOW MANY PRODUCTS WOULD BE SOLD IN **TOTAL** DURING THE FIRST THREE MONTHS OF THE YEAR?

d) SUGGEST ONE CRITICISM OF THIS MODEL

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$N = 64 + 4 \times 1$

$N = 68$

b) 4 IS THE GRADIENT, "m" IN "y=mx+c"

4 IS THE MONTHLY INCREASE
IN THE NUMBER OF PRODUCTS SOLD

c) FIRST 3 MONTHS IN TOTAL, SO m=1,2,3

$m=1,\ N=68$
$m=2,\ N=64+4\times2=72$
$m=3,\ N=64+4\times3=76$

YOU COULD WRITE THESE STRAIGHT DOWN

$\therefore$ TOTAL SALES $= 68 + 72 + 76$
$= 216$

d) THE MODEL SUGGESTS THE NUMBER OF PRODUCTS WOULD CONTINUE TO RISE FOREVER. THIS IS UNLIKELY.

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Examiner Tips and Tricks

- Read – and re-read – the question carefully.Try to "get involved" in the context of the question – however boring it may be!
  - Imagine what happens to a stone as you throw it from a cliff
  - What would it be like to manage a toy factory?
- Draw a diagram – **sketch** a graph of the function being used as the model.
- If you are completely stuck try "doing something" with the maths. For example, if there's a quadratic function – sketch it, factorise it, solve it. Or if there's a polynomial – use f**actor theorem** and **algebraic division** to factorise it.

Worked Example

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